2024A7PS0002G
AI Lab2: Agents

1. Yes, the LLM generated working code in the first attempt after I shared the problem and specifications.

2. - (I used the suggested prompt with images and text from the question pdf.)

3. Breadth-First Search (BFS)

4. BFS is appropriate because every movement in the warehouse has the same cost ie. one grid square per move. This makes BFS achieve the optimal solution.
   Also, it is complete.
   BFS also does not require complex code or a heuristic.

```
Path found!
Positions:
[(1, 1), (1, 2), (1, 3), (1, 4), (2, 4), (2, 5), (2, 6), (2, 7), (1, 7), (1, 8), (1, 9), (1, 10), (1, 11), (1, 12), (1, 13), (1, 14), (1
, 15), (1, 16), (1, 17), (1, 18), (1, 19)]

Actions:
['Right', 'Right', 'Right', 'Down', 'Right', 'Right', 'Right', 'Up', 'Right', 'Right', 'Right', 'Right', 'Right', 'Right', 'Right', 'Rig
ht', 'Right', 'Right', 'Right', 'Right']

Number of moves: 20

Path on warehouse:
####################
#S***.##########G#
#.##############..#
#....##............#
#.######.###.#.###..#
#........#.........#
####################
```